Skill, luck and risk in bits: scoring guesses in feedback-guessing games

The Lexodle project

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Abstract

In a Wordle-like game every guess is a question and the feedback is the answer. We score a guess by the information it gains, in bits. The expected gain H is the Shannon entropy of the way the guess splits the words still possible; *skill* is H as a share of the best real guess's. *Luck* is where the gain you got falls among the gains the guess could have given, a mid-distribution percentile with mean zero. *Risk* is the sign and size of the skew of that distribution about its mean: how far a guess leaned on a rare large gain. Gains add over a game to the bits needed to find the word, so a game's bits split exactly into an expected part and a luck part that is a martingale. A player's habits are read from which of five behaviours each guess is; six playstyles are the pairs of an early and an end habit. The definitions, their properties, the handling of the last try and of several boards, and simulations of the playstyles are given. The maths was written with an AI assistant and has not been peer reviewed.

1 Introduction

In Wordle-like games the player guesses a word. Each letter is marked as in the right place, elsewhere in the word, or absent. After the game two questions arise. Skill: given what was known, how good was the guess? Luck: given the guess, how favourable was the feedback? A score that mixes the two is uninformative.

Entropy-based solvers choose the guess that maximises the expected information in the feedback [1, 6], as for the code game Mastermind [2, 3]. Exact methods give optimal strategies for the original word list [4]; for generalised lists, deciding whether a win within a given number of guesses can be guaranteed is NP-hard [5]. A newspaper review tool reports skill and luck per guess [7]; we know of no published formula. This paper does not address how to play. It defines scores to show a player after the game. The requirements are:

(R1) **Knowledge only.** Skill and risk do not depend on the answer. Luck depends on the answer only through the feedback the player saw.

(R2) **Bounded.** Skill runs from 0 (the guess tells nothing) to 100 (as good as any real guess). Luck is bounded and symmetric about 0. Risk runs from 0 to 100.

(R3) **Fair luck.** Under the law the answer is drawn from, luck has mean zero given the history and is uncorrelated with the skill of the same and earlier guesses (Section 4).

(R4) **One plot.** All three can be read from one plot of the outcomes the guess could have had (Section 8).

(R5) **Accounting.** The bits gained over a game equal the bits expected plus a luck term, exactly, and add up to the bits needed (Section 6).

2 Setting

A game has a finite list of words, and the hidden word is drawn by a *law*: each word w has a chance p_w (equal for every word, or in proportion to how common the word is). Let P be the set of words still possible, with total chance $p(P) = \sum_{w \in P} p_w$. A guess g gives each word $w \in P$ a feedback pattern, and so splits P into groups $G_1, \ldots, G_m$ by pattern. The *chance* of group j is $q_j = p(G_j)/p(P)$, the chance, given what is known, that the answer is in it. The guess is *real* if the game accepts it as a word; it need not be a possible answer. When the law is equal, $q_j = |G_j|/|P|$.

3 Gain and skill

Definition 1. Landing in group j *gains* $I_j = -\log_2 q_j$ bits. The *expected gain* of g is

$$H(g) = \sum_j q_j \, I_j = -\sum_j q_j \log_2 q_j .$$

A bit halves the chance mass still possible: a gain of I_j bits leaves a share 2^{-I_j} of the chance. $H(g)$ is the Shannon entropy [1] of the split. It is 0 exactly when one group holds all the chance (the guess tells nothing), at most $\log_2 m$, and, for equal chances, at most $\log_2 |P|$.

Proposition 1 (Refinement)**.** *If a guess g' splits every group of g further, then $H(g') \geq H(g)$, with equality only if no group is actually divided.*

Proof. Write a group G_j of g as groups of g' with chances q_{jk}, $\sum_k q_{jk} = q_j$. Then $H(g') - H(g) = \sum_j q_j \, H(q_{j1}/q_j, \ldots) \geq 0$, the entropy of the division of each group, which is 0 only for an undivided group. □

Definition 2. Let H^* be the largest $H(g)$ over the real guesses scored. The *skill* of g is $100\, H(g)/H^*$ (and 100 if $H^* = 0$, when no guess tells anything).

Skill depends on P, the law and the guess, never on the answer (R1), and lies in $[0, 100]$ with 100 for the best real guess. The scale is straight in bits: a guess that gains half the bits of the best scores 50. The app scores every word the game accepts; H^* is found by scanning them.

4 Luck

Let I be the gain of a guess when the answer is drawn by the law: it takes the value I_j with chance q_j.

Definition 3. The *luck* of landing in group j is

$$\Lambda_j = \mathbb{P}(I < I_j) - \mathbb{P}(I > I_j) \in (-1, 1),$$

shown $\times 100$. It is $2F_{\text{mid}}(I_j) - 1$, where F_{mid} is the mid-distribution function [8, 9]: the chance of a smaller gain plus half the chance of an equal one.

A larger gain is a luckier outcome. In terms of chances, $\mathbb{P}(I < I_j)$ is the chance mass of the groups more likely than j.

Theorem 2. *Under the law,* $\mathbb{E}[\Lambda] = 0$ *and* $\text{Var}(\Lambda) = (1 - \sum_j q_j^3)/3$. *The luck is strictly between* -1 *and* 1 *and equals* 0 *for every outcome of a guess with one group.*

Proof. $F_{\text{mid}}(I)$ has the distribution of a continuous uniform variable spread over each atom: its mean is $1/2$ and its variance $\frac{1}{12}(1 - \sum q_j^3)$, so $2F_{\text{mid}} - 1$ has mean 0 and variance $(1 - \sum q_j^3)/3$. Ties give the same luck. One group gives $\mathbb{P}(I < I_1) = \mathbb{P}(I > I_1) = 0$. □

Luck in bits. The difference $\lambda_j = I_j - H(g)$ is the gain above the expected gain. Its mean is 0 exactly, and it is what the accounting of Section 6 sums. The percentile Λ is its bounded, scale-free version: Λ and λ have the same sign in the sense that Λ_j is increasing in I_j.

Theorem 3. *Over a game, the luck in bits of successive guesses are martingale differences with respect to the feedback so far; so they are uncorrelated with each other, and with the skill of any guess, which is known before the feedback.*

Proof. Given the history, the pool and the law of the next feedback are fixed and $\mathbb{E}[\lambda_t \mid \text{history}] = 0$ by Theorem 2's argument applied to $I - H$. For $s < t$, $\mathbb{E}[\lambda_s \lambda_t] = \mathbb{E}[\lambda_s \, \mathbb{E}[\lambda_t \mid \mathcal{F}_{t-1}]] = 0$, and skill is a function of the history, so $\mathbb{E}[\text{skill}_t \lambda_t] = 0$. □

The same holds for the percentile Λ, since it too has mean zero given the history; the correlation of the percentiles of different guesses is zero by the same argument.

Under the wrong law. If a game scored luck with equal chances while the answer was drawn by frequency, the mean would no longer be zero. The app scores every game under the law it was drawn by (equal for a daily).

5 Risk

Some guesses gain nearly the same whatever the feedback; others usually gain a little and now and then a lot. Risk is the second kind.

Definition 4. The *raw risk* of g is $D(g) = \mathbb{P}(I < H) - \mathbb{P}(I > H) \in (-1, 1)$: the chance of gaining less than expected less the chance of gaining more. Let D^* be the largest D over the real guesses scored. The *risk* of g is $100 \max(0, D(g))/D^*$, and 0 if $D^* \le 0$.

$D > 0$ means the gain is skewed to the right, the shape of a gamble: a likely small gain, an unlikely large one (mean above median). A guess with equal groups has $D = 0$. Risk is a property of the guess, not of the outcome, so it is known before the feedback and does not move with luck. It is shown beside luck in the review: the chevrons by the risk say whether the outcome landed on the rare large gain or not.

6 Bits over a game

Let guess t land in a group of chance q_t in the pool P_{t-1}, leaving P_t.

Proposition 4 (Chain rule)**.** $\sum_{t\le T} I_t = -\log_2 p(P_T)/p(P_0)$. *With the law normalised,* $\sum_t I_t = -\log_2 p(P_T)$*, which is at most* $-\log_2 p_a$ *for the answer* a*, with equality when* $P_T = \{a\}$.

Proof. $q_t = p(P_t)/p(P_{t-1})$, so $\prod_t q_t = p(P_T)/p(P_0)$, and $a \in P_T$. □

The bits needed to find the word are $-\log_2 p_a$: $\log_2 N$ for equal chances. A guess that wins counts as cutting all the remaining bits. Then, with $\lambda_t = I_t - H_t$,

$$\sum_t I_t = \sum_t H_t + \sum_t \lambda_t,$$

the bits expected (the skill part) plus the luck part, a sum of martingale differences with mean zero (Theorem 3). This is the exact accounting (R5). The review draws it as a waterfall, one bar for each guess, racing to the bits needed.

7 The last try

On the last try nothing is left to learn: only a guess that could be the answer can win. Skill is then the chance the guess wins, as a share of the best chance among the real guesses (the likeliest word); a guess that cannot be the answer scores 0. Luck is the outcome less the chance: $1 - c$ for a win, $-c$ for a miss, where c is the chance of winning; its mean is zero. With one word left, a guess is the word (skill 100, gain 0 bits, risk 0, luck 0, words $1 \to 0$) or a miss (skill 0). Risk is not defined on the last try, nor with one word left.

8 One plot

Draw one column for each possible outcome of the guess, as wide as its chance q_j and as tall as its gain I_j, the outcome that gains least on the left. The area under the columns is H; a dashed line at the mean height marks H, a dotted one the best guess's H^* (skill is the first over the second). The columns taller than the mean are the right tail a risky guess gambles on. A line splits the width in half: the column it crosses has luck 0, and the column landed in is outlined: its distance from the middle is its luck. Skill, risk and luck are then read from one picture (R4).

9 Several boards

In a quartet, one guess serves four boards; in a relay race it serves the board that is on. Each board is scored as a game of its own: its H, its H^*, its luck. A quartet guess has a skill and a luck that are the means over the unsolved boards weighted by the bits still to find on each, $\log_2$ of its words left (at least one bit), so a board with one word left counts for one bit and playing that word is rewarded. Bits add over boards: the gain of a step is the sum of its gains. In a relay race, a board opened later starts with every earlier guess applied; the review reports how many words were left then (its head start).

10 Other modes

The same scores are used in the other single-word modes; only the set of groups or the set of guesses changes.

Lockdown. A green is locked in its place, so a guess must keep every locked letter where it is. The guesses a player could have made are only those words, and H^* and D^* are maxima over them, not over the whole list. Skill is thus measured against the best guess the player was allowed.

Blind. The feedback is two counts, the letters right and in place and the letters right elsewhere, with no tile colours. A group is the set of words that give the same pair of counts, and every score above is taken over those groups. Greens whose places are unknown are not counted as known greens when the phase of a guess is read.

Themed and made lists. The word list is smaller or drawn from a theme, with the chances p_w set by the draw (for places, in proportion to how widely read each is); the scores are those of Section 2 under that law.

A daily in any mode is scored under equal chances. Only classic games, a classic daily included, count towards a playstyle; the other modes are not read for habits.

11 Playstyles

A player's style is read from their last 100 classic games, a classic daily being one. A guess is counted in a phase by the words left n and the greens known: *early* for $n \geq 64$, *end* for $n \leq 16$ or three greens, otherwise the middle, which is not read. Each early guess is counted as the first that applies of:

- a **gambler**: risk ≥ 75;
- a **maximiser**: in the top 10% of the guesses scored, by expected gain;
- an **eliminator**: four or more letters never tried in the game.

and each end guess as the first of:

- a **closer**: the guess fits every clue, so it can win;
- a **prober**: it does not fit, ranks in the top 10%, and three or more words are left.

A *habit* is the behaviour with the largest share of a phase's guesses, if that share is at least 40%. A player is rated once there are ten games and ten guesses in each phase; a habit already held is kept while its share is within 0.05 of the leader's. The *style* is the pair, early habit and end habit:

	prober at the end	closer at the end
maximiser early	analyst	maverick
eliminator early	detective	sweeper
gambler early	wildcard	gambler

Simulated players. Six policies play the real list (English, five letters, 3,000 answers, equal chances), each following its style (the gamblers take their risks on the first two turns and then settle). Over 30 games each, judged game by game with the rules above, the style is read back as follows (script: `experiments/styles/sim2.js`):

policy	read as itself	end habit unread	another style
analyst	20	5	5
maverick	25	5	0
detective	18	3	9
sweeper	24	3	3
wildcard	16	8	6
gambler	26	4	0

A single game has few end guesses, so its end habit is often unread; pooled over the games a player has (the app needs ten) the habits settle. The thresholds come from these simulations, not from measured people, and are the first thing to revisit with real play.

12 Computation

For a pool of n words and c candidate guesses, H and D of every candidate take $O(nc)$ pattern evaluations, grouped by a pattern code; H^* and D^* are maxima over them. The app scans every word the game accepts, in the background while the game is played, one guess after another. When two words are left it scans only those two.

13 Open problems

- Calibrate the playstyle thresholds on real games.
- Is there a closed form for the best achievable H^* for a given pool, beyond the bound $\log_2 m$?
- A skill that rewards the guesses that set up later splits (a two-step H).
- Whether risk predicts the tries a game takes once skill and luck are known.

14 Reproducibility

The definitions are implemented in `app/lib/logic.dart` (gain, H, D, skill, luck, risk, the landscape of all guesses), `app/lib/playstyle.dart` (the behaviours and styles), `app/lib/review_plots.dart` (the plot and the waterfall) and `app/lib/multi_analysis.dart` (several boards; lockdown and blind reuse the same functions with a narrowed guess set or count-based groups), and tested in `app/test`. The playstyle simulation is `docs/paper/experiments/styles`. The scripts of the earlier, pool-size version of the maths (`experiments/*.py`) are kept for the record; they measure the retired expected-pool skill and the retired daring and tries regression, and no claim above rests on them.

References

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